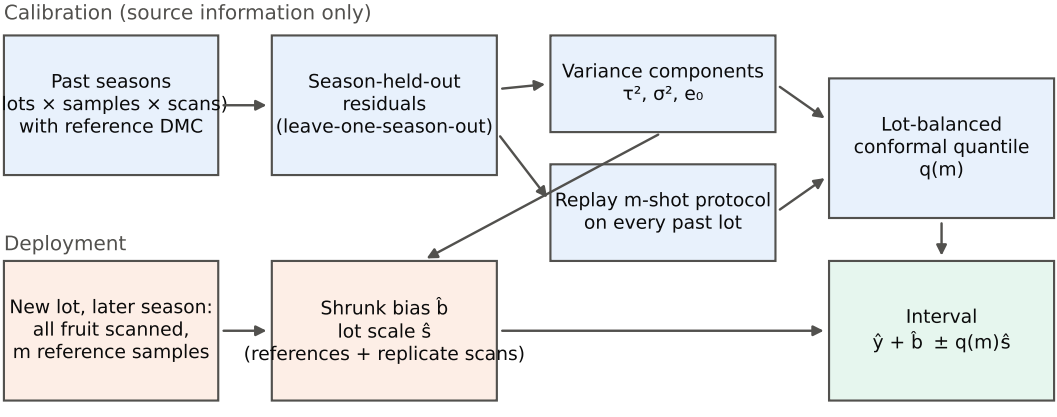
Calibration (source information only)
Past seasons
(lots × samples × scans)
with reference DMC
Season-held-out
residuals
(leave-one-season-out)
Variance components
τ², σ², e₀
Lot-balanced
conformal quantile
q(m)
Replay m-shot protocol
on every past lot
Deployment
New lot, later season:
all fruit scanned,
m reference samples
Shrunk bias b̂
lot scale ŝ
(references + replicate scans)
Interval
ŷ + b̂ ± q(m)ŝ